

手把手教你构建自己的Volcano Job插件

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01

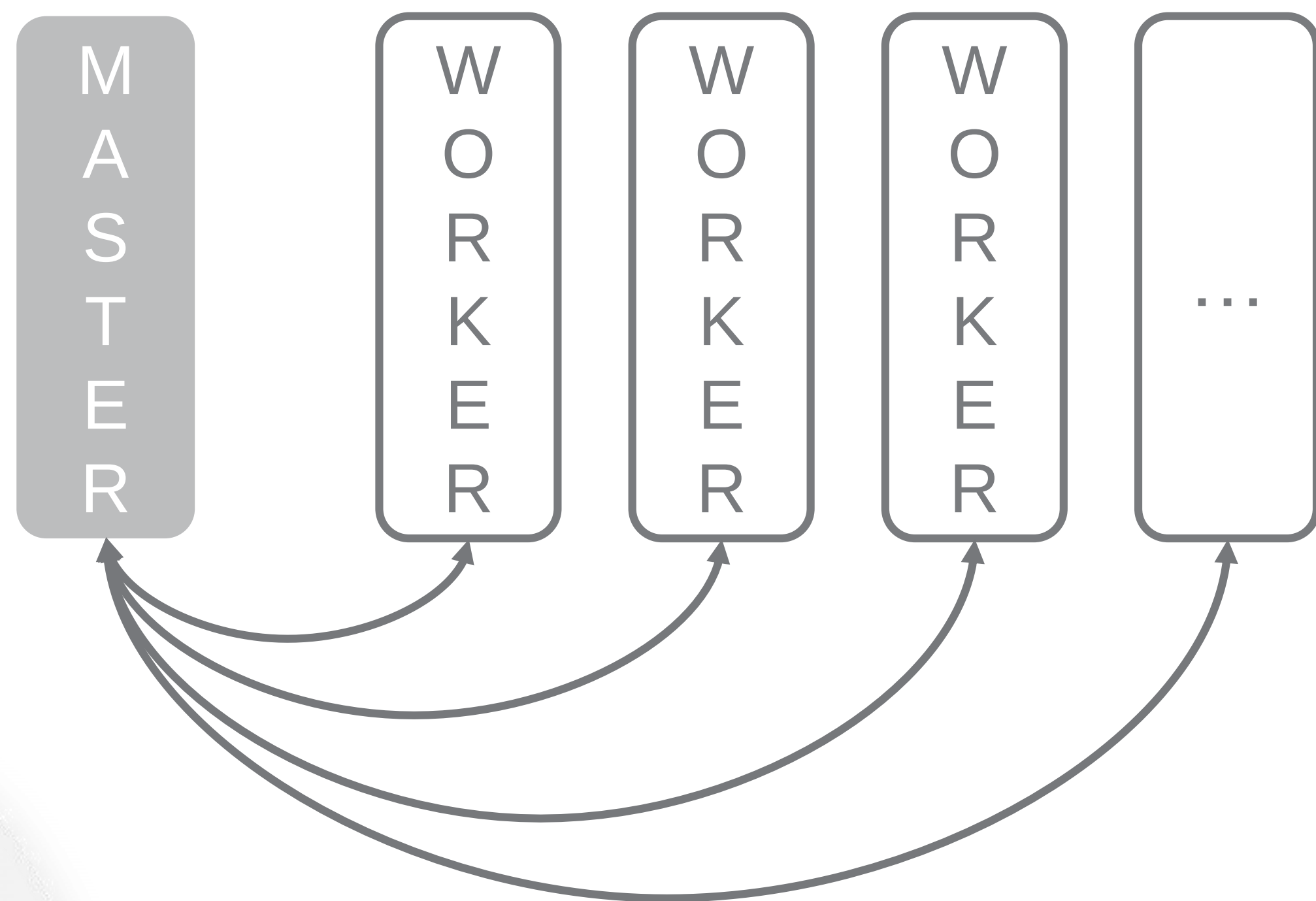
创建一个MPI插件

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什么是MPI



- MPI 是一种用于并行计算机编程的通信协议
- 支持点对点和集体通信
- MPI提供了应用程序消息传递的接口
- MPI是当今高性能计算使用的主要模型

MPI作业调度优化

运行MPI作业的基础要求

为了能成功运行MPI作业，我们至少需实现以下功能

- worker应该先与master启动

- Instance之间通过SSH进行通信，无需密码

- master需要知道worker的地址

- 添加作业的顺序启动能力

- 开启SSH所需的端口

- 启用SSH和SVC插件

- 添加环境变量，记录worker的地址信息

Volcano中可复用的基础能力

ssh 插件

为每个容器配置
ssh免密登录

svc 插件

为master和worker
容器生成地址信息



作业管理生命周期中，插件可自定义修改的关键项



```
type PluginInterface interface {  
    // Name returns the unique name of Plugin.  
    Name() string  
  
    // OnPodCreate is called for all pod when createJobPod  
    OnPodCreate(pod *v1.Pod, job *vcbatch.Job) error  
  
    // OnJobAdd is called when do job initiation  
    // Note: it can be called multi times, must be idempotent  
    OnJobAdd(job *vcbatch.Job) error  
  
    // OnJobDelete is called when killJob  
    // Note: it can be called multi times, must be idempotent  
    OnJobDelete(job *vcbatch.Job) error  
  
    // OnJobUpdate is called when job updated  
    // Note: it can be called multi times, must be idempotent  
    OnJobUpdate(job *vcbatch.Job) error  
}
```


开发实践

编写设计文档

- 设计文档非常重要
- 插件的设计背景和想要添加的功能
- 当前插件的约束限制
- 简单描述插件的实现逻辑



开发实践

初始化

注册 MPI 插件

```
func init() {  
    RegisterPluginBuilder("ssh", ssh.New)  
    RegisterPluginBuilder("env", env.New)  
    RegisterPluginBuilder("svc", svc.New)  
    RegisterPluginBuilder("tensorflow", tensorflow.New)  
    RegisterPluginBuilder("mpi", mpi.New)  
}
```

在pkg/controllers/job/plugins/factory.go/init()中添加registerPluginBuilder函数

编写代码

```
✓ pkg  
  > cli  
  ✓ controllers  
    > apis  
    > cache  
    > framework  
    > garbagecollector  
  ✓ job  
    > helpers  
  ✓ plugins  
    ✓ distributed-framework  
      ✓ mpi  
        go mpi.go
```

在pkg/controllers/job/plugins/distributed-framework/目录下创建mpi目录和mpi.go文件，进行代码编写



开发实践

添加 Job

```
func (mp *mpiPlugin) OnJobAdd(job *batch.Job) error {
    if job.Status.ControlledResources["plugin-"+mp.Name()] == mp.Name() {
        return nil
    }
    job.Status.ControlledResources["plugin-"+mp.Name()] = mp.Name()
    return nil
}
```

将插件信息添加到job的status中

删除 Job

```
func (mp *mpiPlugin) OnJobDelete(job *batch.Job) error {
    if job.Status.ControlledResources["plugin-"+mp.Name()] != mp.Name() {
        return nil
    }
    delete(job.Status.ControlledResources, "plugin-"+mp.Name())
    return nil
}
```

从job的status中删除插件信息



开发实践

创建 Pod

```
func AddDependsOn(job *v1alpha1.Job) {
    mp := controllerMpi.NewInstance(job.Spec.Plugins[controllerMpi.MPIPluginName])
    masterIndex := helpers.GetTaskIndexUnderJob(mp.GetMasterName(), job)
    if masterIndex == -1 {
        klog.Errorln("Failed to find master task")
        return
    }
    if job.Spec.Tasks[masterIndex].DependsOn == nil {
        job.Spec.Tasks[masterIndex].DependsOn = &v1alpha1.DependsOn{
            Name: []string{mp.GetWorkerName()},
        }
    }
}
```

管理启动顺序

```
isMaster := false
workerHosts := ""
env := v1.EnvVar{}
if helpers.GetTaskKey(pod) == mp.masterName {
    workerHosts = mp.generateTaskHosts(job.Spec.Tasks[helpers.GetTaskIndexUnderJob(mp.workerName, job)], job.Name)
    env = v1.EnvVar{
        Name: MpiHost,
        Value: workerHosts,
    }
    isMaster = true
}

// open port for ssh and add MPI_HOST env for master task
for index, ic := range pod.Spec.InitContainers {
    mp.openContainerPort(&ic, index, pod, true)
    if isMaster {
        pod.Spec.InitContainers[index].Env = append(pod.Spec.InitContainers[index].Env, env)
    }
}

for index, c := range pod.Spec.Containers {
    mp.openContainerPort(&c, index, pod, false)
    if isMaster {
        pod.Spec.Containers[index].Env = append(pod.Spec.Containers[index].Env, env)
    }
}
```

通过环境变量记录worker的地址信息

添加启动顺序管理和环境变量信息

```
func (mp *mpiPlugin) openContainerPort(c *v1.Container, index int, pod *v1.Pod, isInitContainer bool) {
    SSHPortRight := false
    for _, p := range c.Ports {
        if p.ContainerPort == int32(mp.port) {
            SSHPortRight = true
            break
        }
    }
    if !SSHPortRight {
        sshPort := v1.ContainerPort{
            Name:      "mpijob-port",
            ContainerPort: int32(mp.port),
        }
        if isInitContainer {
            pod.Spec.InitContainers[index].Ports = append(pod.Spec.InitContainers[index].Ports, sshPort)
        } else {
            pod.Spec.Containers[index].Ports = append(pod.Spec.Containers[index].Ports, sshPort)
        }
    }
}
```

打开SSH端口

为master和worker容器打开SSH端口



开发实践

在webhook中添加插件信息

```
if _, ok := job.Spec.Plugins["mpi"]; ok {  
    if _, ok := plugins["svc"]; !ok {  
        plugins["svc"] = []string{}  
    }  
  
    if _, ok := plugins["ssh"]; !ok {  
        plugins["ssh"] = []string{}  
    }  
}
```

为启用了mpi插件的作业强制启用ssh和svc插件，如果没有ssh和svc插件，mpi作业是无法正常运行的

02

测试你的代码

UT & E2E

Unit testing

1. 考虑需要测试的场景
2. 在 `mpi.go` 同目录创建 `mpi_test.go` 文件
3. 编写测试代码
4. 运行测试用例

E2E testing

1. 考虑需要测试的场景
2. 在 `test/e2e` 目录编写测试代码
3. 运行测试用例



03

提交至 Volcano 社区

在 GitHub 创建 Pull Request

为插件创建并编写设计文档

设计文档被采纳后，在Volcano社区创建 Pull Request 提交代码

等待自动化构建运行通过

等待社区 reviewer 检视代码

检视通过后，您的代码将会合并到 Volcano 仓库

add mpi plugin #2194

Merged volcano-sh-bot merged 1 commit into volcano-sh:master from hwdef:add-mpi-job-plugin yesterday

Conversation 53 Commits 1 Checks 12 Files changed 10

hwdef commented 21 days ago Member

ref: #1806
ref: #1874

👍 1 🚀 1

volcano-sh / volcano Public

Unwatch 76 Fork 515 Starred 2.3k

< Code Issues 148 Pull requests 35 Actions Projects Wiki Security Insights

Filters is:pr is:open Labels 64 Milestones 7 New pull request

	Author	Label	Projects	Milestones	Reviews	Assignee	Sort
add shuffle action • size/L							3
#2174 opened 3 days ago by Thor-wl							
support elastic × size/S							1
#2173 opened 5 days ago by qiankunli							
add overhead for pod request × size/M							9
#2170 opened 5 days ago by jiangxiaobin96							
fix controller job cache not sync latest version issue • size/L							3
#2169 opened 5 days ago by wpeng102							
vGPU support for volcano (design doc) × size/L							3



加入Volcano



Website: <https://volcano.sh/en/>



Github: <https://github.com/volcano-sh/volcano>



Slack Channel: <https://volcano-sh.slack.com/>



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